

Class: XII 'Science'

Subject: Physics

F.M: 75

P.M: 30

Attempt all questions:

Group A

Tick the best alternative for the following questions:

[11x1=11]

- A charged particle is released from rest in a region of steady and uniform magnetic and electric fields which are parallel to each other. The particle will move in a
a) straight line b) circle c) helix d) cycloid
- A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the center of the coil is B. It is then bent into a circular loop of 'N' turns. The magnetic field at the center of the coil will be
a) NB b) N^2B c) 2NB d) $2N^2B$
- A fly wheel of moment of inertia 4kgm^2 rotating about an axis with 120 revolutions/min is brought to rest in 5 secs. The torque needed is
a) 5Nm b) 10Nm c) 20Nm d) 40Nm
- A current loop placed in a non-uniform magnetic field experience
a) a force of repulsion b) a force of attraction
c) a torque but not force d) a force and a torque
- The amplitude of a simple harmonic motion is 'a'. The displacement of a particle when its energy is half K.E and half P.E is
a) $\frac{a}{2}$ b) $\frac{a}{\sqrt{2}}$ c) $\frac{a}{4}$ d) $\frac{a}{8}$
- If the earth suddenly contracts to half of its present radius, then the duration of day is
a) 12 hours b) 8 hours c) 6 hours d) 4 hours
- Two sound waves are represented by; $y_1 = a \sin (wt - kx)$ and $y_2 = b \cos (wt - kx)$. The phase difference between the two waves is
a) π b) $\frac{\pi}{4}$ c) $\frac{3\pi}{4}$ d) $\frac{\pi}{2}$
- The ratio of specific charge of a β - radiation to that of electron is
a) 1:1 b) 1:2 c) 2:1 d) 4:1
- Electrons execute 1.4×10^6 revolutions/sec in a magnetic field of flux density
a) $4 \times 10^{-5} \text{ T}$ b) $5 \times 10^{-5} \text{ T}$ c) $6 \times 10^{-5} \text{ T}$ d) $8 \times 10^{-5} \text{ T}$
- Which of the following is not deflected by electric and magnetic field?
a) α -particle b) β -particle c) photon d) proton

11. Due to the flow of current in a circular loop of radius 'R' the magnetic induction produced at the center of the loop is 'B'. The magnetic moment of the loop is

- a) $\frac{2\pi BR^3}{\mu_0}$ b) $\frac{\mu_0}{2\pi BR^2}$ c) $\frac{BR^2}{2\pi\mu_0}$ d) $\frac{B^2 R}{2\pi\mu_0}$

Group B

Answer the following questions:

1. a) What is meant by moment of inertia of an object about an axis? State the factors on which moment of inertia of a rigid body depend on. [8x5=40]
b) Find the moment of inertia of a thin rod with mass 'M' and length 'L' about an axis passing through its center of mass and perpendicular to the rod. [1+1]

[3]

OR

- a) Is the statement "the bob of a simple pendulum moves faster at the lowest position for larger amplitude" true? Justify your answer? [2]
b) A spring balance has a scale that reads from 0 to 50 kg. The length of the scale is 20cm. A body suspended from this spring, when displaced and released, oscillates with a period of 0.60 s. What is the weight of the body? [3]
2. a) A proton moving in a straight line enters a strong magnetic field along the field direction. How will its path and speed change? [2]
b) A positive charge of $1.5 \mu\text{C}$ is moving with a speed of $2 \times 10^6 \text{ m/s}$ along positive x-axis. A magnetic field, $\vec{B} = (0.2\hat{j} + 0.4\hat{k})$ Tesla acts in the space. Find the vector form and magnitude of the force acting on the charge. [3]
3. a) State and explain Biot-Savart Law. [2]
b) Derive an expression for magnetic field strength at the center of a current carrying circular conductor. [3]
4. a) Describe Millikan's oil drop experiment to determine the charge of an electron. [4]
b) How did this experiment lead to the quantum nature of electric charge? [1]
5. a) What is the strong argument for the particle nature of cathode rays? [1]
b) An electron beam after being accelerated from rest through a potential difference of 5000 volts in vacuum is allowed to impinge normally on a fixed surface. If the incident current is $50 \mu\text{A}$, determine the force exerted on the surface assuming that it brings the electrons to rest. [4]

OR

6. The mass of a proton is 1847 times that of an electron. An electron and a proton moving with same speed are projected in a uniform magnetic field. Compare the radii of their circular path? [2]

b) An electron moving with velocity 2×10^7 m/s describes a circle of magnetic field of strength 0.02 T. If the specific charge $\left(\frac{e}{m}\right)$ of electron is 1.76×10^{11} C/kg, find the approximate diameter of the circular path?

6. a) What is a travelling wave? [3]

b) Derive an equation for the travelling wave. [1]

c) You have a semi-infinite string holding it at one end and move the end of the string up and down in a sinusoidal motion in time at a rate of 50 times per second. The distance from equilibrium that you move the end is 0.05 meters. The speed of the wave on the string is 200 m/s. Find an equation for $y(x, t)$ that describes the displacement of the string from equilibrium in terms of x and t (the distance from the end) and time. [2]

7. a) A body is moving in a circular path with constant speed. Is this motion a simple harmonic? Why? [1]

b) Find an expression for the total energy of a particle in simple harmonic motion. Also, sketch the variation of K.E and P.E of the particle with respect to displacement. [4]

8. a) At what conditions, the torque produced in a rectangular coil placed in a magnetic field is maximum and minimum? [1]

b) A uniform magnetic field of 100 Gauss is in east to west direction and you are given 44 cm long wire with a current carrying capacity of at least 10A. What is the shape and orientation of the loop made of this wire which yields the maximum turning effect on the loop? What will be the magnitude of the maximum torque? [4]

Group C

Answer the following questions:

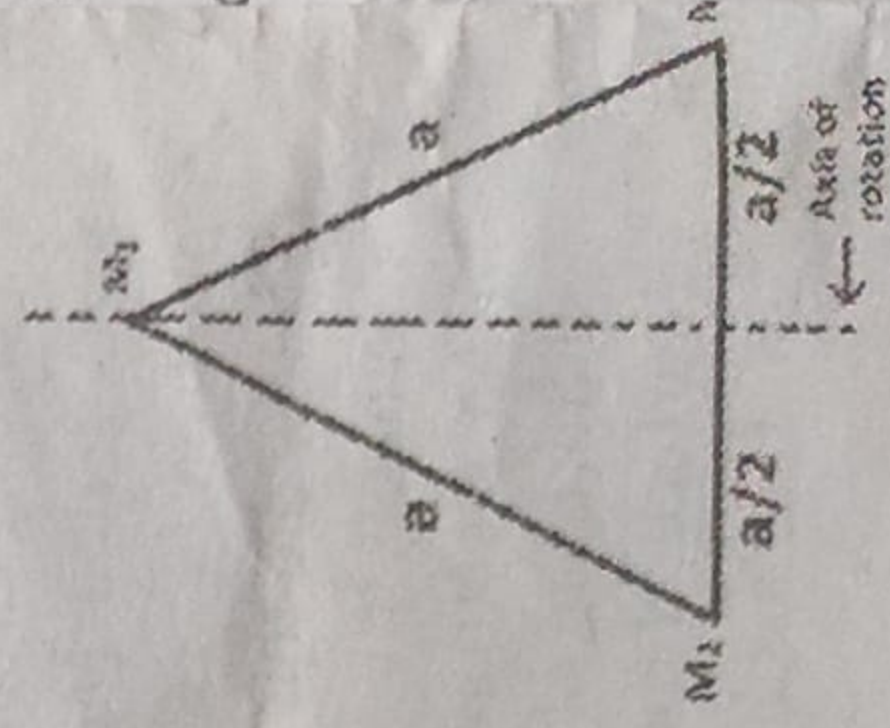
9. a) Why is a ladder more apt to slip when you are high up on it than when you just begin to climb? [2]

b) Three-point masses M_1 , M_2 and M_3 are located at the vertices of an equilateral triangle of length 'a' as shown in figure. Find the moment of inertia of the system about an axis along the altitude of the triangle passing through M_1 . [2]

c) State and explain the principle of conservation of angular momentum for a system of objects. [2]

d) On a rotating merry-go-round, one child sits on a horse near the outer edge and another child sits on a lion halfway out from the center.

i) Which child has the greater linear velocity? [1]



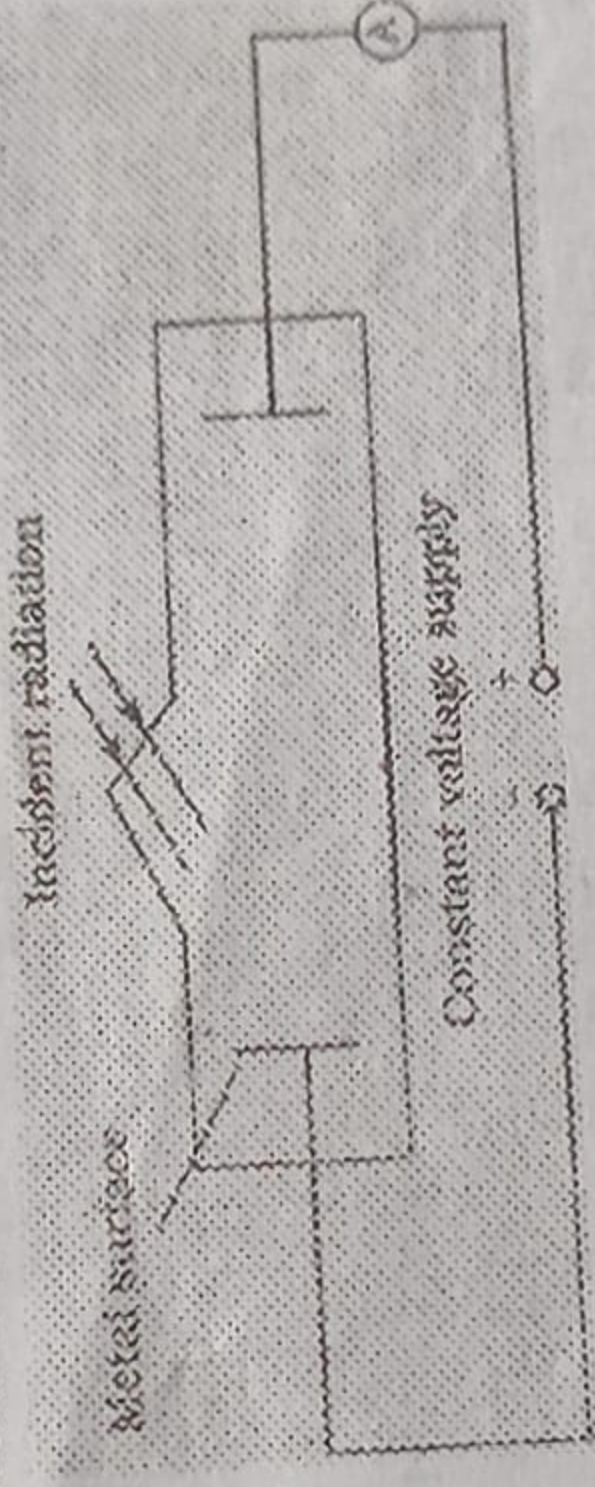
ii) Which child has the greater angular velocity? Explain. [1]

10. a) Describe J. J Thomson's experiment for the determination of specific charge of an electron. [4]

b) An electron emitted by a heated cathode and accelerated through a potential difference of 2kV, enters a region with uniform magnetic field of 0.15T. Determine the trajectory of the electron if the field (i) is transverse to its initial velocity, (ii) makes an angle of 30° with the initial velocity. [4]

OR

The apparatus shown below is used to investigate photoelectric emission from a metal surface when electromagnetic radiation is shone on the surface. The intensity and frequency of the incident radiation can be varied as required.



i) Explain what is [2]

meant by photoelectric emission from a metal.

ii) What is the name given to the minimum frequency of the radiation that produces a current in the circuit? [1]

iii) A particular source of radiation produces a current in the circuit. Explain why the current in the circuit increases as the intensity of the incident radiation increases. [2]

iv) Discuss the quantum theory of radiation to explain energy and momentum of photon. And, discuss the characteristics properties of photon. [3]

photon. [1]

11. a) Explain the concept of Hall effect. Hence find out expression for Hall voltage. [4]

b) Discuss the mechanism of Hall Effect and hence find out expression for Hall voltage. [4]

c) The current sensitivity of a galvanometer increases by 20% when its resistance is increased by a factor 2. Calculate the factor by which the voltage sensitivity changes? [3]

BEST OF LUCK

Group A

1. How many odd numbers of three different digits can be formed from the integers 1, 2, 3, 4, 5? [1X11=11]

- a) 24 b) 36 c) 12 d) 48

2. The value of $P(15, 2)$ is:

- a) 200 b) 205 c) 210 d) 105

3. In how many ways can 3 letters be posted in 4 letter boxes?

- a) 64 b) 81 c) 12 d) 36

4. The co-ordinates of the point where the line through the points A(5, 6, 1) and (5, 1, 6) crosses the XY plane is:

- a) (5, 6, 0) b) (7, 5, 0) c) (5, 0, 7) d) (5, 7, 0)

5. The length of the latus rectum of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is:

- a) $\frac{32}{5}$ b) $\frac{16}{5}$ c) $\frac{5}{32}$ d) $\frac{64}{5}$

6. The centre of the ellipse $9x^2 + 4y^2 - 18x - 16y - 11 = 0$ is:

- a) (-1, -2) b) (1, 2) c) (2, 1) d) (-2, -1)

7. The length of conjugate axis of the hyperbola $\frac{(x-1)^2}{16} - \frac{(y+3)^2}{9} = 1$ is:

- a) 8 b) 4 c) 6 d) 3

8. The co-ordinates of the foci of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = -1$ is:

- a) $(\pm 5, 0)$ b) $(0, \pm 5)$ c) $(0, \pm 4)$ d) $(\pm 4, 0)$

9. The function $f(x) = |x-1|$ is:

- a) Continuous at $x=1$
b) Differentiable at $x=1$
c) Not differentiable at $x=1$
d) Both a & c

10. $\frac{d}{dx}(2^{x^2}) = ?$

- a) $2^{x^2} \cdot 2x$ b) $2^{x^2} \cdot 2x \cdot \ln 2$ c) $2^{x^2} \ln x$ d) $2^{x^2} \cdot 2x \cdot \ln x$

11. $\frac{d}{dx}(\sec h^{-1}x) = ?$

a) $-\frac{1}{\sqrt{1-x^2}}$

b) $\frac{1}{x\sqrt{1-x^2}}$

c) $\frac{1}{x\sqrt{1-x^2}}$

d) $\frac{1}{x\sqrt{x^2-1}}$

Group B

[8x5=40]

1. In how many ways can the letters of the word 'CALCULUS' be arranged so that the two C's do not come together?
2. A person has got 12 acquaintances of whom 8 are relatives. In how many ways can he invite 7 guests so that 5 of them may be relatives?
3. Find the seventh term and the middle term in the expansion of: $(2x+y)^{12}$
4. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_n x^n$, then Prove that:
 $C_1 - 2C_2 + 3C_3 - \dots + n(-1)^{n-1}C_n = 0$.
5. If the roots of the equation $x^2 - 2cx + ab = 0$ be real and unequal, then prove that the roots of $x^2 - 2(a+b)x + a^2 + b^2 + 2c^2 = 0$ will be imaginary.
6. Find the eccentricity, the co-ordinates of the vertices, foci and length of the latus rectum of the conic $x^2 + 4y^2 - 4x + 24y + 24 = 0$.
7. Find the derivative of: i) $x^y \cdot y^x = 1$ ii) $X^{\sin h^2(x/a)}$
8. Evaluate (Using L' Hospital's Rule):

ii) $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$

i) $\lim_{x \rightarrow 0} \frac{xe^x - \ln(x+1)}{x^2}$

[8x3=24]

Group C

1. Find the derivative from first principle of: i) x^x ii)

2. $\ln \sin^{-1}(x)$ [4+4]

3. State Rolle's Theorem. Interpret it geometrically verify

Rolle's Theorem for $f(x) = x(x-1)^2$ in $[0,1]$. Discuss the applicability

[1+1+4+2]

of Rolle's Theorem for $f(x) = \frac{1}{x^2-1}$ in $[-2, 2]$

[4+4]

4. i) Find the angle of intersection of the curves $y=x^2$ and $x=y^2$.

- ii) If α and β are the roots of $Px^2 + qx + q = 0$, then prove that:

$$\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{q}{p}} = 0$$

~All the Best~



KATHMANDU NOVEL SECONDARY SCHOOL



First Term Examination 2078

Class: XII 'Science'

Subject: Biology [Botany+ Zoology]

F.M: 75

P.M: 30

Subject:

Group A

Circle the correct one from the given alternatives.

(11x1=11)

1. The tissue where cells are thin-walled, living, isodiametric along with intercellular space is

- a) Meristem b) Sclerenchyma c) Parenchyma d) Collenchyma

2. "Stele" includes

- a) Vascular tissue
b) Pith + vascular tissue
c) Pericycle + Pith + vascular tissue
d) Endodermis + Pericycle + Pith + vascular tissue

3. Which condition of xylem is shown in the figure?

- a. Exarch b. Endarch c. Mesarch d. Centarch

4. DNA sequence ATCGGACT has complementary sequence

- a. TAGCCTGT b. TAGCCTCT c. ATCGGACT d. TAGCCTGA

5. tRNA attaches amino acid at its

- a. 3' end of acceptor arm b. 5' end of acceptor arm
c. anticodon loop d. extra loop

6. Genes are responsible for the production of

- a) Carbohydrates b) Fats c) Protein d) vitamins

7. The substance that fills the bone cavity is

- a. Osteoclasts b. Chondroblasts c. Collagen d. Bone marrow

8. Myelin of the nerve fibers of the CNS is produced and maintained by

- a. Astrocytes b. Microglia c. Schwann cells d. Oligodendrocytes

9. Which one is the wrong match among the following:

- | | |
|-------------------------|-------------------------------|
| a. Glandular epithelium | Salivary gland |
| b. Squamous epithelium | Skin of frog |
| c. Ciliated epithelium | Columnar epithelium |
| d. Columnar epithelium | Peritoneum of the body cavity |

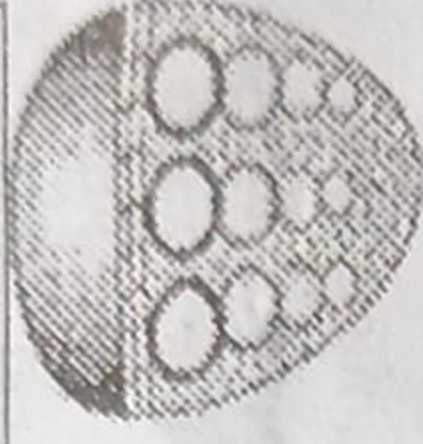
10. Which one of the following correctly matches the dental formula of a normal adult human?

- a. 2123/2103 b. 2133/2133 c. 3123/3123 d. 2123/2123

11. Which cells of Crypts of Lieberkuhn secrete antibacterial lysozyme?

- a. Chief cells b. Kupffer cells c. Paneth cells d. Mucus cells

Toward periphery



Toward Pith

TCP M

Group B

SHORT QUESTION:

[8x5=40]

1. What is the name of tissue which are in continuous state of active cell division? Mention the structure and functions of such type of tissue. (1+2+2)
2. What is complex tissue? Describe the components of xylem. Also mention how xylem differ from phloem (1+2+2)
3. What is vascular bundle? Describe various types of vascular bundle with necessary figure and example. (1+4)
4. Define nucleic acid. How nucleotide differ from nucleoside? Describe different components of nucleotides.
5. What is peristalsis. Describe the process of digestion of protein in the alimentary canal of human. (1+4)
- ✓ 6. Draw a well labelled diagram of a typical myelinated neuron. (5)
7. Differentiate between the bone and cartilage. (5)
8. Write a short note on types of simple epithelial tissue. (5)

Group C

LONG QUESTION:

[3x8=24]

1. Describe the internal structure of dicot root with suitable diagram. How does it differ from monocot root? (3+2+3)
2. What is DNA replication? Describe the mechanism of semi- conservative mode of DNA replication with necessary figure. (1+5+2)
3. Define digestion. Describe the alimentary canal of the human in detail. [8]

BEST OF LUCK



Class: XII 'Science'

Time: 3 hours

Subject: Compulsory English

F.M: 75

P.M: 30

1. Read the following text about the importance of interdependence at the present time and do the given tasks. -15

We've never been as self-conscious about ourselves as we seem to be these days. The popular magazines are filled with advice on things to do with a self: how to find it, identify it, nurture it, protect it, and even, for special occasions, weekends, how to lose it transiently. There are instructive books, best sellers on self-realization, self-help, and self-development. Groups of self-respecting people pay large fees for three day sessions together, learning self-awareness. Self-enlightenment can be taught in college electives.

You'd think, to read about it, that we'd only just now discovered selves. Having long suspected that there was something alive in there, winning the place, separate from everyone else, absolutely individual and independent, we've celebrated by giving it a real name.

Myself.

It is an interesting word, formed long ago in much more social ambiguity than you'd expect. The original root was *se* or *seu*, simply the pronoun of the third person, and most of the descendant words, except "self" itself, were constructed to allude to other, somehow connected people; "sibs" and "gossips," relatives and close acquaintances, came from *seu*. *Se* was also used to indicate some side or apart, hence words like "separate," "secret" and "segregate." From an extended root *swedh* it moved into Greek meaning people of one's own sort, and *ethos*, meaning the customs of such people. "Ethics" means the behavior of people like one's self, one's own ethics

We tend to think of ourselves as the only wholly unique in nature, but it is not so. Uniqueness is so commonplace a property of living things that there is really nothing at all unique about it. The phenomenon can't be unique and universal at the same time. Even individual, free-swimming bacteria can be viewed as unique entities distinguishable from each other even when they are the progeny of a single clone. Spudich and Koshland have recently reported that motile microorganisms of the same species are like solitary eccentrics in their swimming behavior. When they are searching for food, some tumble in one direction for precisely so many seconds before quitting, while others tumble differently and for different, but characteristic, periods of time. If you watch them closely, tethered by their flagellae to the surface of an antibody-coated slide, you can tell them from each other by the way they twirl, as accurately as though they had different names.

Beans carry self-labels, and are marked by these as distinctly as a mouse by his special smell. The labels are glycoproteins, the lectins, and may have something to do with negotiating the intimate and essential attachment between the bean and the nitrogen fixing bacteria which live as part of the plant's flesh embedded in root nodules. The lectin from one line of legume has a special affinity for the surfaces of the particular bacteria which colonize that line, but not for bacteria from other types of bean. The system seems designed for the maintenance of exclusive partnerships. Nature is pieced together by little snobberies like this.

Corals are biologically self-conscious. If you place polyp of the same genetic line together, touching each other, they will fuse and become a single polyp, but if the lines are different, one will reject the other. Fish can tell each other apart as individuals, by the smell of self. So can mice, and here the olfactory discrimination is governed by the same H2 locus which contains the genes for immunologic self-marking.

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Beans carry self-labels, and are marked by these as distinctly as a mouse by his special smell. The labels are glycoproteins, the lectins, and may have something to do with negotiating the intimate and essential attachment between the bean and the nitrogen fixing bacteria which live as part of the plant's flesh, embedded in root nodules. The lectin from one line of legume has a special affinity for the surfaces of the particular bacteria which colonize that line, but not for bacteria from other types of bean. The system seems designed for the maintenance of exclusive partnerships. Nature is pieced together by little snobberies like this.

Coral polyps are biologically self-conscious. If you place polyp of the same genetic line together, touching each other, they will fuse and become a single polyp, but if the lines are different, one will reject the other. Fish can tell each other apart as individuals, by the smell of self. So can mice, and here the olfactory discrimination is governed by the same H2 locus which contains the genes for immunologic self-marking.

The only living units that seem to have no sense of privacy at all are the nucleated cells that have been detached from the parent organism and isolated in a laboratory dish. Given the opportunity, under the right conditions, two cells from wildly different sources, a yeast cell, say, and a chicken erythrocyte, will touch, fuse, and the two nuclei will then fuse as well, and the new hybrid cell will now divide into monstrous progeny. Naked cells, lacking self-respect, do not seem to have any sense of self. The markers of self, and the sensing mechanisms responsible for detecting such markers, are conventionally regarded as mechanisms for maintaining individuality for its own sake, enabling one kind of creature to defend and protect itself against all the rest. Selfness, seen thus, is for self-preservation. In real life, though, it doesn't seem to work this way. The self-marking of invertebrate animals in the sea, who must have perfected the business long before evolution got around to us, was set up in order to permit creatures of one kind to locate others, not for predation but to set up symbiotic households. The anemones who live on the shells of crabs are precisely finicky; so are the crabs. Only a single species of anemone will find its way to only a single species of crab. They sense each other exquisitely, and live together as though made for each other.

Sometimes there is such a mix-up about selfness that two creatures, each attracted by the molecular configuration of the other, incorporate the two selves to make a single organism. The best story I've ever heard about this is the tale told of the nudibranch and medusa living in the Bay of Naples. When first observed, the nudibranch, a common sea slug, was found to have a tiny vestigial parasite, in the form of a jellyfish, permanently affixed to the ventral surface near the mouth. In curiosity to learn how the medusa got there, some marine biologists began searching the local waters for earlier developmental forms, and discovered something amazing. The attached parasite, although apparently so specialized as to have given up living for itself, can still produce offspring, for they are found in abundance at certain seasons of the year. They drift through the upper waters, grow up nicely and astonishingly, and finally become full-grown, handsome, normal jellyfish. Meanwhile, the snail produces snail larvae, and these too begin to grow normally, but not for long. While still extremely small, they become entrapped in the tentacles of the medusa and then engulfed within the umbrella-shaped body.

At first glance, you'd believe the medusae are now the predators, paying back for earlier humiliations, and the snails the prey. But no. Soon the snails, undigested and insatiable, begin to eat, browsing away first at the radial canals, then the borders of the rim, finally the tentacles, until the jellyfish becomes reduced in substance by being eaten while the snail grows correspondingly in size. At the end, the arrangement is back to the first scene, with a full-grown nudibranch basking, and nothing left of the jellyfish except the round, successfully edited parasite, safely affixed to the skin near the mouth.

It is a confusing tale to sort out, and even more confusing to think about. Both creatures are designed for this encounter, marked as selves so that they can find each other in the waters of the Bay of Naples. The collaboration, if you want to call it that, is entirely specific; it is only this species of medusa and only this kind of nudibranch that can come together and live this way. And, more surprising, they cannot live in any other way; they depend for their survival on each other. They are not really selves, they are specific others.

What does the collaboration of the selves tell us about our identity?

The thought of these creatures gives me an odd feeling. They do not remind me of anything, really. I've never heard of such a cycle before. They are bizarre, that's it, unique. And at the same time, like a vaguely remembered dream, they remind me of the whole earth at once. I cannot get my mind to stay still and think it through.

A. Read the text and fill in the gaps with the best alternative.

[10x1=10]

- a. Persons who are related to you and who live after you, such as your child or grandchild are called
 i. relatives ii. descendants iii. Siblings
- b. '.....' means to turn or spin around and around quickly
 i. Twirl ii. Fold iii. Fetter

● ○ REDMI NOTE 9 PRO MAX i. Polysaccharide chains (glycans) attached to amino acid side-chains
 ○ ○ GLAXO CAMERA ii. Enzymes iii. Meat products

- d. are carbohydrate-binding proteins that are highly specific for sugar groups that are part of other molecules and so cause agglutination of particular cells.
 i. Lactose ii. Legumes iii. Lectins
- e. are abnormal tissue growths that most often look like small, flat bumps or tiny mushroom like stalks found in the phylum Cnidaria and the medusa
 i. Polyps ii. Buds iii. Tulips
- f. is a biological interaction where one organism, the predator, kills and eats another organism, its prey.
 i. Murder ii. Homicide iii. Predation
- g. is a genus of flowering plants in the buttercup family.
 i. Anemone ii. Ammonia iii. Amoeba
- h. A/An person is extremely or excessively particular, exacting, or meticulous in taste or standards.
 i. arrogant ii. finicky iii. Symbiotic
- i. is a group of soft-bodied, marine gastropod molluscs which shed their shells after their larval stage.
 i. Shell ii. Nudibranch iii. Jellyfish
- j. A is a free-swimming sexual form of a coelenterate such as a jellyfish, typically having an umbrella-shaped body with stinging tentacles around the edge.
 i. medusa ii. sea horse iii. sea slug

B. Answer the following questions.

[5x1=5]

- What are the indicators of the fact that we are very self-conscious about ourselves these days?
- How do fish recognize each other?
- What is the function of individuality?
- What does the mix-up of two selves tell us about our identity?
- Why is the author disturbed by the thought of the creatures like the nudibranch and medusa?

2. Write short answers to the following questions.

[5x2=10]

- What, according to *The Earnest Young Woman*, are the indications that her husband has fallen in love with Famous Actress? (*A Matter of Husband*)
- Why did Oliver Sacks become so fascinated by Hook? (*On Libraries*)
- What do you understand by the line 'The hills untied their bonnets'? (*A Day*)
- Why is Gaston surprised with his wife's expression towards the end of the story? (*A Respectable Woman*)
- How did the neighbors respond to the woman's pregnancy? (*Neighbors*)

3. Write long answers to the following questions.

[2x5=10]

- The story shows that linguistic and cultural barriers do not create any obstacle in human relationship. Justify. (*Neighbors*)
- How does Famous Actress make a fool of Earnest Young Woman? (*A Matter of Husband*)

4. Write a news story based on the given information.

[7]

University opened - University of Information Technology - opened among a big crowd at Patan - built from the grant financial support of China - project started on 2011 - ended in 2017 - has the capacity for 200 students a year - best wishes from the minister of education.

- Write an email to your foreign friend describing the impact of corona virus in your country. [8]
- Suppose you are the General Manager of Buddha Airlines. Issue a press release on behalf of the airlines about the cancellation of flights to the mountain regions due to the poor weather condition. [15]

[10×1=10]

7. Do as indicated in brackets and rewrite the sentences.

- There's little we can do about it,? (Add a question tag)
- I'd say that's enough,? (Add a question tag)
- Let us go,? (Add a question tag)
- I had it when I left the office so I (mustn't / must/should) have lost it on the way to home.
(Choose one form from the bracket to complete the sentence)
- It (can't/might/must) have got lost in the post. These things happen sometimes. (Choose one form from the bracket to complete the sentence)
- He is an extrovert. He has got (much/many/few/little) friends. (Choose one form from the bracket to complete the sentence)
- Leap years occur (each/ every) four years. (Choose one form from the bracket to complete the sentence)
- Your garden is superb. (all /most / none /both/ some) of the flowers are beautiful. (Choose one form from the bracket to complete the sentence)
- The teacher and poet (is/are) coming. (Choose one form from the bracket to complete the sentence)
- He is poor, but he is very kind and helpful. (Rewrite the sentence using 'despite')

[5×1=5]

8. Choose the best options.

- The word 'shine' has the vowel sound.....
i. ei/
ii. /ai/
- The word 'freight' has the vowel sound.....
i) /ei/
ii) /ai/
- The relationship between the breastfeeding woman and her nonbiological nursing infant is called
i. sibling-in-law
ii. milk kinship
iii. half- sibling
iv. step- sibling
- The prefix 'hyper' in the word hyperloop means.....
i. fixed
ii. above
iii. again
iv. angry
- In the sport of cricket, is a run scored by the batting team if the batsman has not hit the ball with their bat, but the ball has hit the batsman's body or protective gear.
i. googly
ii. leg bye
iii. yorker
iv. wicket

